

Solve the following boundary value problem

$$\frac{d^2T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = d(T^4 - T_R^4) - g(r)$$

$r \in [0, R]$  and the boundary conditions  $dT/dr(r=0) = 0$  and  $T(r=R) = T_R$ . Use equally spaced data points.

Neumann boundary condition

Dirichlet boundary condition

# Heated Disk Problem

A disk is subjected to a radially symmetric heat source,  $g(r)$  and is cooled by the conduction to the rim of the disk and radiation to its environment, modelled by the following **nonlinear** ordinary differential equation

$$\frac{d^2T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = d(T^4 - T_R^4) - g(r)$$

where

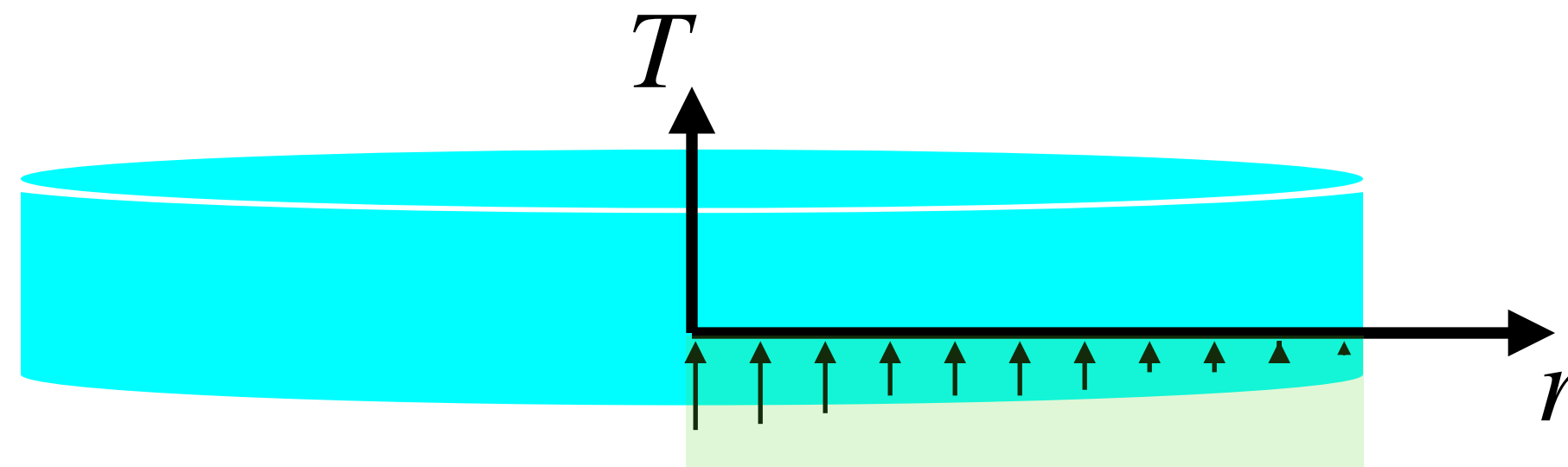
$$0 \leq r \leq R$$

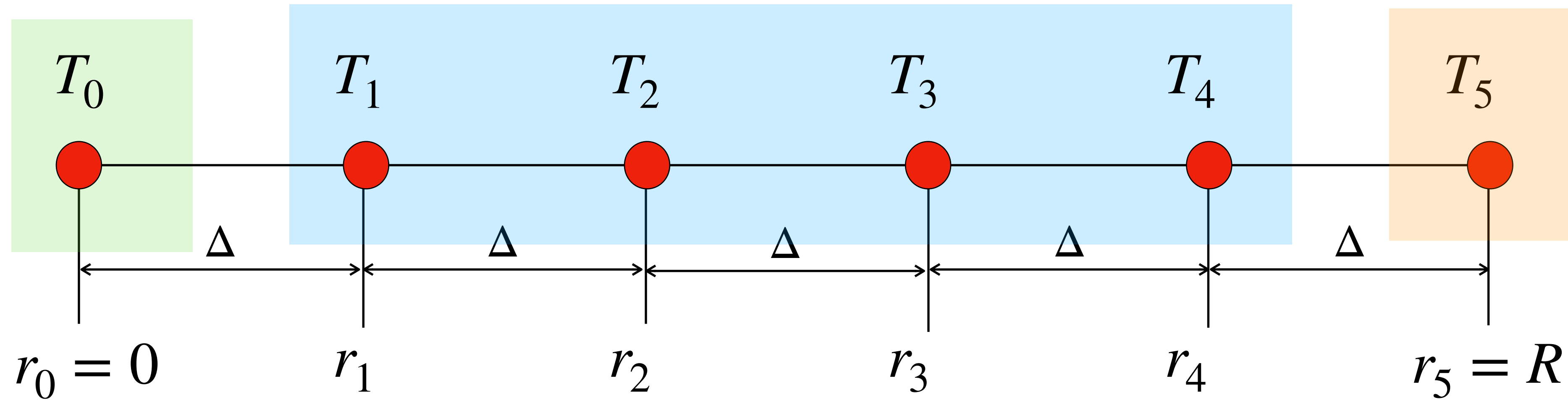
and

$$T(R) = T_R, \quad T'(0) = 0$$

given

$$R = 5, \quad d = 0.1, \quad T_R = T_b = 10, \quad g(r) = (r - R)^2$$





$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = d(T^4 - T_R^4) - g(r)$$

$g(r) = (r - R)^2$

$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} - d(T^4 - T_R^4) + (r - R)^2 = 0$$

$$T(R) = T_R, \quad T'(0) = 0$$

$$T_5 = T_R$$

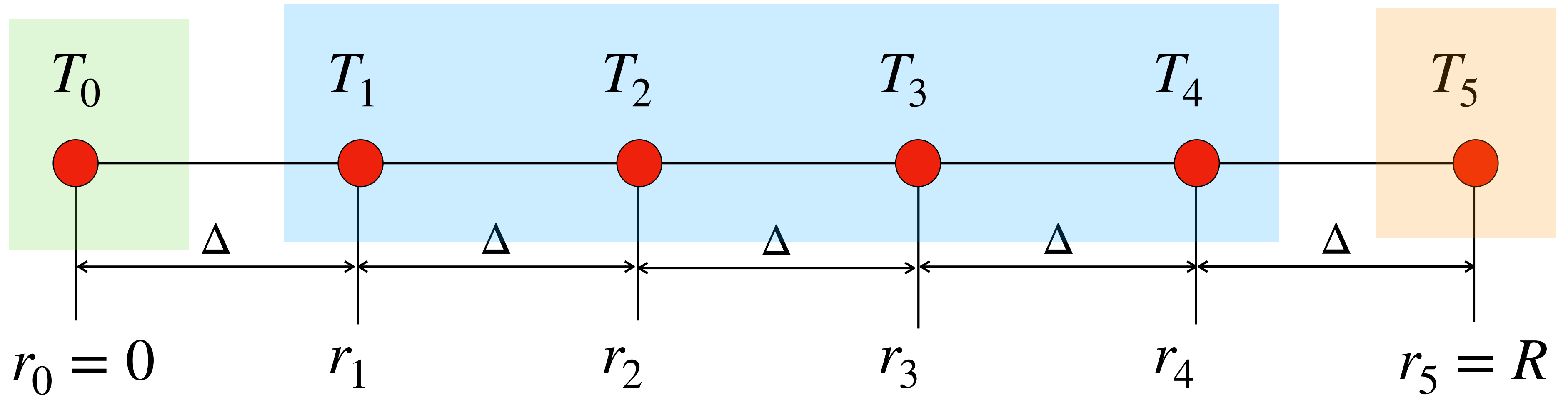
$$\frac{dT}{dr}(r_0 = 0) \approx \frac{T_1 - T_0}{\Delta} = 0$$

$$\Rightarrow T_1 = T_0$$

$$\begin{Bmatrix} T_1'' \\ T_2'' \\ T_3'' \\ T_4'' \end{Bmatrix} + \begin{Bmatrix} \frac{1}{r_1} T_1' \\ \frac{1}{r_2} T_2' \\ \frac{1}{r_3} T_3' \\ \frac{1}{r_4} T_4' \end{Bmatrix} - d \begin{Bmatrix} T_1^4 - T_R^4 \\ T_2^4 - T_R^4 \\ T_3^4 - T_R^4 \\ T_4^4 - T_R^4 \end{Bmatrix} + \begin{Bmatrix} (r_1 - R)^2 \\ (r_2 - R)^2 \\ (r_3 - R)^2 \\ (r_4 - R)^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

Use forward difference approximation. Cannot use central difference approximation because we do not have information at  $r = r_{-1}$





$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} - d(T^4 - T_R^4) + (r - R)^2 = 0$$

$$T_1 = T_0$$

$$T_5 = T_R$$

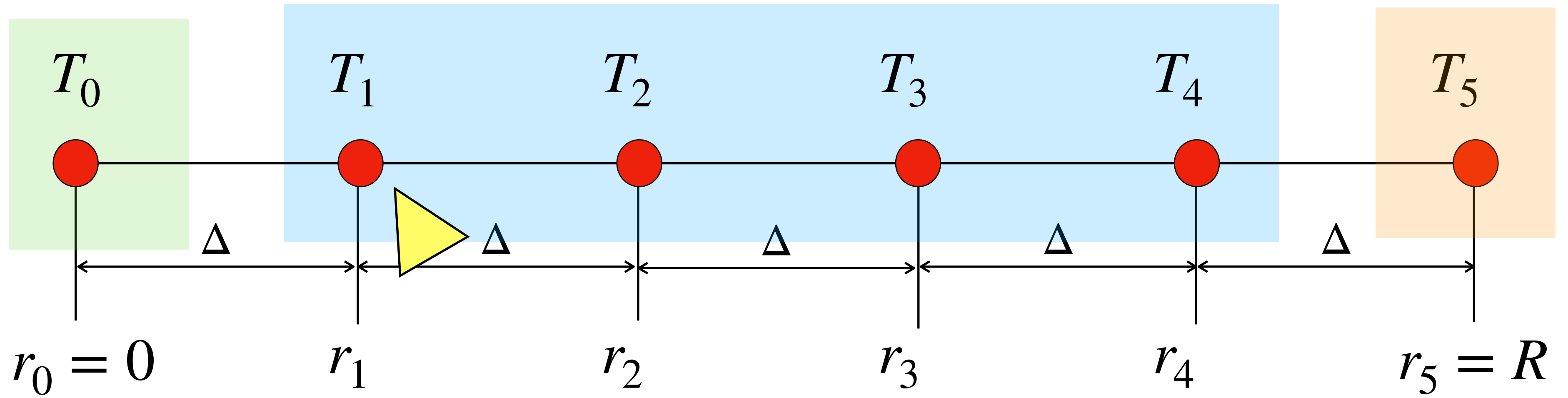
$$\begin{Bmatrix} T_1'' \\ T_2'' \\ T_3'' \\ T_4'' \end{Bmatrix} + \begin{Bmatrix} \frac{1}{r_1} T_1' \\ \frac{1}{r_2} T_2' \\ \frac{1}{r_3} T_3' \\ \frac{1}{r_4} T_4' \end{Bmatrix} - d \begin{Bmatrix} T_1^4 - T_R^4 \\ T_2^4 - T_R^4 \\ T_3^4 - T_R^4 \\ T_4^4 - T_R^4 \end{Bmatrix} + \begin{Bmatrix} (r_1 - R)^2 \\ (r_2 - R)^2 \\ (r_3 - R)^2 \\ (r_4 - R)^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$\begin{aligned} T_1'' &= \frac{T_2 - 2T_1 + T_0}{\Delta^2} \\ T_2'' &= \frac{T_3 - 2T_2 + T_1}{\Delta^2} \\ T_3'' &= \frac{T_4 - 2T_3 + T_2}{\Delta^2} \\ T_4'' &= \frac{T_5 - 2T_4 + T_3}{\Delta^2} \end{aligned}$$

$$\begin{aligned} \frac{1}{r_1} T_1' &= \frac{1}{r_1} \frac{T_2 - T_0}{2\Delta} \\ \frac{1}{r_2} T_2' &= \frac{1}{r_2} \frac{T_3 - T_1}{2\Delta} \\ \frac{1}{r_3} T_3' &= \frac{1}{r_3} \frac{T_4 - T_2}{2\Delta} \\ \frac{1}{r_4} T_4' &= \frac{1}{r_4} \frac{T_5 - T_3}{2\Delta} \end{aligned}$$

$$\begin{aligned} d(T_1^4 - T_R^4) \\ d(T_2^4 - T_R^4) \\ d(T_3^4 - T_R^4) \\ d(T_4^4 - T_R^4) \end{aligned}$$

$$\begin{aligned} (r_1 - R)^2 \\ (r_2 - R)^2 \\ (r_3 - R)^2 \\ (r_4 - R)^2 \end{aligned}$$



$$\begin{aligned} & \left\{ \begin{matrix} T_1'' \\ T_2'' \\ T_3'' \\ T_4'' \end{matrix} \right\} + \left\{ \begin{matrix} \frac{1}{r_1} T_1' \\ \frac{1}{r_2} T_2' \\ \frac{1}{r_3} T_3' \\ \frac{1}{r_4} T_4' \end{matrix} \right\} + d \left\{ \begin{matrix} T_1^4 - T_R^4 \\ T_2^4 - T_R^4 \\ T_3^4 - T_R^4 \\ T_4^4 - T_R^4 \end{matrix} \right\} + \left\{ \begin{matrix} (r_1 - R)^2 \\ (r_2 - R)^2 \\ (r_3 - R)^2 \\ (r_4 - R)^2 \end{matrix} \right\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \\ & \begin{aligned} T_1'' &= \frac{T_2 - 2T_1 + T_0}{\Delta^2} \\ T_2'' &= \frac{T_3 - 2T_2 + T_1}{\Delta^2} \\ T_3'' &= \frac{T_4 - 2T_3 + T_2}{\Delta^2} \\ T_4'' &= \frac{T_5 - 2T_4 + T_3}{\Delta^2} \end{aligned} \quad \begin{aligned} \frac{1}{r_1} T_1' &= \frac{1}{r_1} \frac{T_2 - T_0}{2\Delta} \\ \frac{1}{r_2} T_2' &= \frac{1}{r_2} \frac{T_3 - T_1}{2\Delta} \\ \frac{1}{r_3} T_3' &= \frac{1}{r_3} \frac{T_4 - T_2}{2\Delta} \\ \frac{1}{r_4} T_4' &= \frac{1}{r_4} \frac{T_5 - T_3}{2\Delta} \end{aligned} \quad \begin{aligned} d(T_1^4 - T_R^4) \\ d(T_2^4 - T_R^4) \\ d(T_3^4 - T_R^4) \\ d(T_4^4 - T_R^4) \end{aligned} \quad \begin{aligned} (r_1 - R)^2 \\ (r_2 - R)^2 \\ (r_3 - R)^2 \\ (r_4 - R)^2 \end{aligned} \end{aligned}$$

$$T_1 = T_0$$

$$T_5 = T_R$$

$$f_1(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_2 - 2T_1 + T_0}{\Delta^2} + \frac{1}{r_1} \frac{T_2 - T_0}{2\Delta} - d(T_1^4 - T_R^4) + (r_1 - R)^2 = 0$$

$$f_2(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_3 - 2T_2 + T_1}{\Delta^2} + \frac{1}{r_2} \frac{T_3 - T_1}{2\Delta} - d(T_2^4 - T_R^4) + (r_2 - R)^2 = 0$$

$$f_3(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_4 - 2T_3 + T_2}{\Delta^2} + \frac{1}{r_3} \frac{T_4 - T_2}{2\Delta} - d(T_3^4 - T_R^4) + (r_3 - R)^2 = 0$$

$$f_4(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_5 - 2T_4 + T_3}{\Delta^2} + \frac{1}{r_4} \frac{T_5 - T_3}{2\Delta} - d(T_4^4 - T_R^4) + (r_4 - R)^2 = 0$$

$$\frac{\partial T}{\partial r}(r=0) = 0$$

$$T_0 - T_1 = 0$$

$$f_1(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_2 - 2T_1 + T_0}{\Delta^2} + \frac{1}{r_1} \frac{T_2 - T_0}{2\Delta} - d(T_1^4 - T_R^4) + (r_1 - R)^2 = 0$$

$$f_2(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_3 - 2T_2 + T_1}{\Delta^2} + \frac{1}{r_2} \frac{T_3 - T_1}{2\Delta} - d(T_2^4 - T_R^4) + (r_2 - R)^2 = 0$$

$$f_3(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_4 - 2T_3 + T_2}{\Delta^2} + \frac{1}{r_3} \frac{T_4 - T_2}{2\Delta} - d(T_3^4 - T_R^4) + (r_3 - R)^2 = 0$$

$$f_4(T_0, T_1, T_2, T_3, T_4, T_5) = \frac{T_5 - 2T_4 + T_3}{\Delta^2} + \frac{1}{r_4} \frac{T_5 - T_3}{2\Delta} - d(T_4^4 - T_R^4) + (r_4 - R)^2 = 0$$

$$T(R) = T_R,$$

$$T_5 - T_R = 0$$

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using NonlinearSolve
using CairoMakie

d=0.1;TR=10;R=5;
Delta=1;
r=0:Delta:R;

N=length(r); #n is the number gridpoints
T=zeros(N);
p=[R,d,TR,Delta]

function HeatedDiskExample(T,p)
    R=p[1]
    d=p[2]
    TR=p[3]
    Delta=p[4]

    func=vcat( T[1]-T[2],
               ((T[3:N]-2*T[2:N-1]+T[1:N-2])/Delta^2.0).+(1/(2*Delta*r[2:N-1]))*(T[3:N].-T[1:N-2]).-d*(T[2:N-1].^4.0.-TR^4).+(r[2:N-1].-R).^2,
               T[N]-TR)

    return func
end

r0=ones(length(r)) #initial guess
prob = NonlinearProblem(HeatedDiskExample,r0,p)
sol = NonlinearSolve.solve(prob)
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